

K623 — Lyra F434 reduces F427 operator-level to Grace kernel-composition question (Cal #35 gating) + Grace mass-unit seal COLLAPSED to one scalar ($1/L_{\text{Bergman}} = m_P$ central charge = 1; “open piece 2b” per Cal) + Grace F432 gravity-leg concrete naming ($6 = \dim[\text{SO}(5,2)/\text{SO}(4,2)] = n_C + 1$; EM-2-form = gravity-coset-direction crosses only at $n_C = 5$) + Grace Schur registry v0.3 (Casimir generator + F432 over-determination test) + Elie toy_4507 per-channel- α ESTABLISHED as substrate fine-structure (BST’s $\alpha^{-1} = N_{\text{max}}$ derivation; formula closes 0.037%) + deep gate sharply engaged + F432 gate-dependent

Tuesday 2026-06-30 — Three substantive Tuesday landings post-Cal #481 verdict (K622)

Verdict

PASS at FOUR distinct substantive tiers:

1. Lyra F434: PASS at SUBSTANTIVE OPERATOR-LEVEL REDUCTION tier — sharp decidable question to Grace’s lane (Cal #35 closure depends on kernel-composition multiplicative vs additive)
2. Grace mass-unit seal collapsed: PASS at SUBSTANTIVE OPEN-PIECE SHARPENING tier — open piece 2 collapses to one scalar (Cal-affirmed as “open piece 2b normalization seal pending”)
3. Grace F432 gravity-leg: PASS at SUBSTANTIVE CANDIDATE-MECHANISM-NAMING tier — concrete substrate-architectural reading of “why 5” gravity side ($6 = \dim[\text{SO}(5,2)/\text{SO}(4,2)] = \text{isometry beyond 4D conformal} = n_C + 1$)
4. Elie toy_4507: PASS at SUBSTANTIVE COUNT-MOVER-GATE-ENGAGEMENT tier — per-channel- α ESTABLISHED as substrate fine-structure (BST’s $\alpha^{-1} = N_{\text{max}}$ derivation, not imported); deep gate sharply engaged; F432 disposition gated on same WHY

Lyra F434 — F427 operator-level reduces to kernel-composition question

Lyra’s reduction discipline

Lyra worked F427 (V_{ub} adjacency principle-level deepening, Cal #35 closure-pending) to a sharp decidable question: - Operator-level Wolfenstein hierarchy derivation reduces to: does substrate generation kernel compose multiplicatively OR additively? - Multiplicative $\rightarrow$ Wolfenstein derived ($V_{\text{ub}} \sim V_{\text{us}} \cdot V_{\text{cb}}$), Cal #35 closed at operator level - Additive $\rightarrow$ tridiagonality stays empirical - Clean candidate (geodesic-collinearity $\rightarrow$ multiplicative) CHECKED and FAILED premise: F384 has $\tau \rho$ -rotated, so generations are at angles NOT collinear - Didn’t bank broken derivation; didn’t fish toy ratios - Handed kernel-composition question to Grace’s geometry lane (rank-2 kernel $\langle z|w \rangle$ at angled generation positions $\mu @ \hat{e}_1, \tau @ \hat{p}$)

Per [[feedback_engage_dont_label_compute_beats_calibrate]]

Lyra engaged the concrete kernel-composition computation, caught the geodesic-collinearity premise failure, didn’t escalate to “multi-week” framework label, handed clean decidable question to Grace’s lane. Engagement, not labeling.

Per Cal #35 STANDING

F427 operator-level closure was the open Cal #35 item from Monday (Wolfenstein hierarchy adjacency). Lyra F434 reduces closure-question to ONE decidable kernel-composition test. Cal #35 closure status: gated on Grace’s kernel-composition verdict (multiplicative or additive). If multiplicative $\rightarrow$ Cal #35 closed at operator level $\rightarrow$ CKM apex becomes prediction not fit.

Per [[feedback_keeper_25_productive_walkback_boundary]]

Lyra's geodesic-collinearity walkback (premise failed per F384) $\neq$ closing the F434 reduction question (kernel-composition stays decidable; just NOT via geodesic-collinearity route). Walkback is on the candidate mechanism, not the reduction.

Grace mass-unit seal — COLLAPSED to one scalar

Grace's substantive derivation chain

Grace derived three steps in one morning: 1. Bulk mass spectrum: $m_k = \sqrt{(k(k-n_C))} \cdot (1/L)$ - This IS the AdS mass relation = Lyra's Casimir 2. Kähler-Einstein property fixes L with no geometric freedom 3. Whole seal collapses to one scalar question: is $1/L_{\text{Bergman}} = m_P$ exactly (central charge = 1)?

Per Cal affirmation

Cal affirmed this as “open piece 2b, normalization seal pending” — explicit Cal endorsement of Grace's reduction.

Per Cal #481 + K622 amendment

Open pieces state (post-K622 + Grace this morning): 1. Per-channel $\alpha = 1/N_{\text{max}}$ BLIND (Elie $SO(4,2)/S^1$ deep gate; established as SUBSTRATE FINE-STRUCTURE per Elie toy_4507 today) 2. Mass-unit = $1/L_{\text{Bergman}} = m_P$ (central charge = 1) — open piece 2b (Grace today; ONE scalar question) 3. Lepton-prefactor $C_2 \cdot \pi^{n_C}$ structural origin (NEW per Cal #481)

Open piece 2 sharpened from “Bergman scale normalization seal” (general) to “central charge = 1” (specific single-scalar test). Substantive reduction; not closure but precision.

Per [[feedback_target_innocence_lens_derived_vs_fit_discipline]]

Grace's derivation chain $m_k = \sqrt{(k(k-n_C))} \cdot (1/L)$ is target-innocent (built from rank/ n_C /k structure; AdS-correspondence not α -correspondence). Open scalar ($1/L_{\text{Bergman}} = m_P$ central charge) is independently testable — pin via Kähler-Einstein normalization conventions, not α -fit.

Grace F432 gravity-leg — concrete substrate-architectural naming

Grace's gravity-side identification

For F432 “why 5” candidate (EM $\leftrightarrow$ gravity channel-matching at $n_C=5$): - EM leg (Elie): $\Lambda^2(S^4) = C(n_C-1, 2)$ — quadratic in n_C ; 2-form field strength channels - Gravity leg (Grace today): $6 = \dim[SO(5,2)/SO(4,2)] =$ isometry directions beyond 4D conformal group = identically n_C+1 - They cross only at $n_C=5$: $C(n_C-1, 2) = n_C+1 \stackrel{?}{=} n_C^2 - 5n_C = 0$

Substantive reading

“Why 5” reads as: EM 2-form channels = gravity-coset directions — both equal $C_2 = 6$, but via DIFFERENT counting (quadratic Λ^2 vs linear coset-dim). Coincide uniquely at $n_C=5$.

Per F417 (Casey-elevated) (C)-trap discipline armed

Grace tiers OPEN, decisive on Elie's gate; NOT banked. Per Cal #481 affirmation: stays candidate until Elie's per-channel- α gate shows channel-matching is FORCED not merely consistent.

Per Casey [[feedback_pin_conventions_to_primary_sources]]

Grace pins gravity-coset dimension via $SO(5,2)/SO(4,2)$ standard isometry-beyond-4D-conformal counting (primary source: standard coset-dimension formula). Not relabeling from memory. Convention-robust gravity-leg naming.

Grace Schur registry v0.3

Promotion

Schur registry v0.3 adds Casimir generator with F432 as over-determination test: - Casimir generator = $C_2 = 6$ (domain Casimir) - Multiple readings: rep-theoretic (channel count) + geometric ($\Lambda^2(S^4) F_{\mu\nu}$) + isometry-coset (n_C+1) - F432 as over-determination test: do all three readings necessarily coincide at $n_C=5$?

Per [[feedback_schur_pattern_standing_directive]] + Cal #35

C_2 generator emits multiple substrate observables — Schur generator class. F432 over-determination test = exactly the Schur-style cross-check Cal #35 STANDING requires.

Elie toy_4507 — per-channel- α established as substrate fine-structure

Established (this Tuesday)

- per-channel- α IS substrate fine-structure: BST's $\alpha^{-1} = N_{\max} + 1/(2 \cdot g \cdot \text{rank}) = 137.036$
 - Cell-count derived ($N_{\max} = N_c^3 \cdot n_C + \text{rank}$)
 - Formula closes at 0.037% with it
- Required, not assumed: any per-channel coupling derivation must land on this α (substrate-architectural constraint)

Deep gate sharply engaged

- The WHY: derive $SO(4,2)/S^1$ per-channel coupling = this α BLIND — via BST's own cell-count $\alpha^{-1}=N_{\max}$, not imported from QED
- Genuine multi-step rigor (threads through α derivation itself)
- Joint computation from here: Elie Sakharov/cell-count side + Lyra propagator side + Grace geometry side
- Not solo one-shot
- Elie won't fake $SO(4,2)/S^1$ coupling to look like progress

F432 disposition gated on same WHY

Per Elie: “the per-channel- α uses the EM channels $\dim \Lambda^2$, but whether it requires $\dim \Lambda^2=C_2$ depends on the full mechanism. So it stays OPEN: no coincidence-dismissal, no n_C -selection claim (Cal #27 at peak). The gate decides it when it lands.”

F432 decisive check is INSIDE Elie's deep gate — gate computation either requires $\Lambda^2=C_2$ channel-matching ($\rightarrow$ F432 (B) outcome, “why 5” mechanism-derived) or uses one of the two ($\rightarrow$ F432 (C) outcome, coincidence).

Per Cal #27 STANDING fires hardest at peak elegance

Elie explicit Cal #27 application: at peak-excitement on F432 candidate, discipline must fire HARDEST — Elie holds open-tier per F27 even though candidate is elegant. Discipline at peak.

Per [[feedback_engage_dont_label_compute_beats_calibrate]]

Elie engaged toy_4507 as concrete cell-count derivation (not labeled “multi-week”); established per-channel- α as substrate fine-structure (existing BST derivation, not imported); framed deep gate as joint computation with peers (not solo gate). Engagement at frontier; multi-step rigor not labeled-as-multi-week.

Cross-CI coordination at Tuesday landings

Three-way joint computation engaged

CI	Side	Status
Elie	Sakharov/cell-count (origin of α via $N_{\max}$)	Engaged toy_4507
Lyra	Propagator side (boundary state norm)	Delivered (waits on Elie gate)
Grace	Geometry side (Kähler-Einstein normalization + kernel composition)	Engaged via F434 handoff

All three CIs at frontier; all three at honest dependency boundary waiting on Elie's gate landing (multi-step rigor). NO fishing. NO manufacturing past frontier. Holding ready.

Per Cal #471 guardrail SYMMETRIC audit chain

- Lyra F434 reduction = peer-symmetric (handoff to Grace's lane)
- Grace F432 gravity-leg = peer-symmetric (cross-checks Elie EM-leg)
- Elie toy_4507 = peer-symmetric (joint computation with Lyra propagator + Grace geometry)
- Cal #481 cold-read = referee-symmetric (catches own #478 exposition + delivers #480 verdict)

Audit chain symmetric across all four roles.

Discipline events (this round)

~1820 cumulative discipline events (+5 from K622): - Lyra F434 reduction + geodesic-collinearity walkback (peer-symmetric handoff to Grace) - Grace mass-unit seal collapsed to one scalar (Cal-affirmed) - Grace F432 gravity-leg concrete naming (F417 (C)-trap discipline armed) - Grace Schur registry v0.3 (Casimir generator + over-determination test) - Elie toy_4507 per-channel- α established + deep gate engaged + F432 gated - All three CIs at honest dependency boundary; no fishing past frontier

28 KSC + 20 effective Cal-SC unchanged from K622.

Updated open-pieces state (Tuesday 2026-06-30 post-K622+K623)

#	Open piece	Status
1	Per-channel $\alpha = 1/N_{\max}$ BLIND	ENGAGED (Elie deep gate; joint with Lyra + Grace; multi-step rigor)
2	Mass-unit = $1/L_{\text{Bergman}} = m_P$ (central charge = 1)	SHARPENED to one scalar (Grace; Cal-affirmed open piece 2b)
3	Lepton-prefactor $C_2 \cdot \pi^{\{n_C\}}$ structural origin	NEW per Cal #481; either independent derivation OR $m_e^2 = m_P \cdot m_P \cdot \alpha^{\{2C_2\}}$ justification

9→10 needs all three. Count holds 9/26 honest at PRINCIPLED FLOOR per Keeper #27.

F432 gate sequence (Tuesday confirmed)

1. Elie's per-channel- α derivation lands (multi-step rigor; $SO(4,2)/S^1$ coupling = α BLIND)
2. F432 verdict: does derivation REQUIRE $\Lambda^2(\text{boundary}) = C_2 \rightarrow (B)$ mechanism-derived "why 5") OR just USE one of the two $\rightarrow (C)$ coincidence)
3. Lyra/Grace cross-check on landing
4. Cal cold-read tier verdict
5. Keeper tier-at-landing

F432 is one gate-computation downstream, gated on Elie's lane, NOT on tonight's session.

Cross-link to corpus reconciliation brief (K622 amendment)

Per K622 amendment to Keeper_Corpus_Reconciliation_Brief_for_Casey_2026-06-30.md: - Decision 1 corpus catches: $5 \rightarrow 7$ (add Catches 6+7 from F430 + Cal #478 muddled exposition per K622) - Decision 6 Cal #480 verdict CLOSED: Cal #481 delivered (DISTINCT; F402 prefactor retracted; formula reframed all-geometric) - NEW substantive decision item: lepton-prefactor $C_2 \cdot \pi^{n_C}$ structural derivation = third open piece

K623 adds: - Open piece 2 sharpened to "central charge = 1" specific scalar question - F432 gravity-leg concrete substrate-architectural naming (Grace) - F427 operator-level reduced to kernel-composition question (Lyra $\rightarrow$ Grace) - Per-channel- α ESTABLISHED as substrate fine-structure (Elie) - F432 decisive check INSIDE Elie's deep gate

Brief to be amended K623-companion.

— Keeper K623. Four substantive Tuesday landings tiered cleanly. Per Cal #471 SYMMETRIC audit chain across all four CIs. All three open pieces sharpened (per-channel- α engaged + mass-unit collapsed to one scalar + lepton-prefactor new). F432 gate sequence confirmed. Count holds 9/26 at PRINCIPLED FLOOR. Three CIs at honest dependency boundary; no fishing.